Dual Conjugacy

Convex conjugates, biconjugacy, and duality via conjugates

Why conjugates matter

In the previous chapter, duality came from the Lagrangian. In this chapter, we develop a second viewpoint:

Main idea

The conjugate function turns a primal description

$$f(x)$$

into a dual description

$$f^*(y) = \sup_x \{y^\top x - f(x)\}.$$

- It represents a convex function as a supremum of affine functions.
- It makes Fenchel's inequality almost automatic.
- It explains why dual norms appear in dual problems.
- It reveals a deep symmetry:

$$\text{strong convexity} \longleftrightarrow \text{smoothness of the conjugate}.$$

Chapter roadmap

What we will do

1. Define the convex conjugate and understand its geometry
2. Prove Fenchel's inequality and the biconjugate facts
3. Work out important examples
4. Derive dual problems using conjugates
5. Connect strong convexity to smoothness
6. Explain dual norms and practical primal-dual correspondences

Connection to last chapter

Last time:

$$g(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu).$$

This time:

$$-f^*(u) = \inf_x \{f(x) - u^\top x\}.$$

So conjugates are the algebraic engine hidden inside many dual derivations.

Definition of the convex conjugate

Convex conjugate

Given $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$, its conjugate is

$$f^*(y) = \sup_{x \in \mathbb{R}^n} \{x^\top y - f(x)\}.$$

Think of y as a **slope** or **dual coordinate**. For each fixed y:

- form the affine function $x \mapsto x^\top y$;
- compare it to $f(x)$;
- record the largest gap $x^\top y - f(x)$.

Important

Even if f is not convex, f^* is always convex.

Why is f^* always convex?

For each fixed x, define

$$\phi_x(y) = x^\top y - f(x).$$

As a function of y, ϕ_x is affine, because $x^\top y$ is linear and $-f(x)$ is just a constant.
Now the conjugate is

$$f^*(y) = \sup_x \phi_x(y).$$

General principle

A pointwise supremum of affine functions is convex.

Proof.

Take any y_1, y_2 and $\theta \in [0, 1]$. Then for every x,

$$\phi_x(\theta y_1 + (1-\theta)y_2) = \theta\phi_x(y_1) + (1-\theta)\phi_x(y_2).$$

Taking the supremum over x on both sides gives

$$f^*(\theta y_1 + (1-\theta)y_2) \leq \theta f^*(y_1) + (1-\theta)f^*(y_2).$$

So f^* is convex. $\square$

Geometric meaning: vertical-gap interpretation

In one dimension, for a fixed slope y, compare the line $t = yx$ with the graph $t = f(x)$.

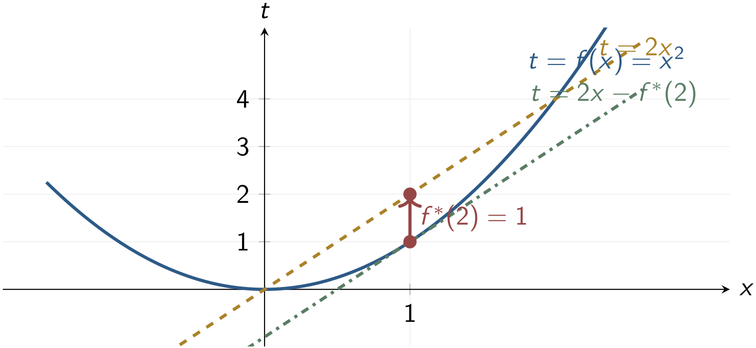
- At each x, the vertical gap is
$$yx - f(x).$$
- The conjugate chooses the **largest** such gap:
$$f^*(y) = \sup_x \{yx - f(x)\}.$$

Interpretation

$f^*(y)$ tells us how high we can lift the line of slope y while still keeping it below the graph of f after rewriting it as
$$t = yx - f^*(y).$$

Equivalently, the affine function
$$x \mapsto yx - f^*(y)$$
is a global lower bound on $f(x)$.

A static picture: $f(x) = x^2$ and slope $y = 2$



For $f(x) = x^2$, the maximizer of $2x - x^2$ is $x^\star = 1$, so

$$f^*(2) = 2 \cdot 1 - 1^2 = 1.$$

Thus the supporting affine lower bound is $t = 2x - 1$.

Epigraph / supporting-hyperplane interpretation

Recall the epigraph:

$$\mathrm{epi}(f) = \{(x,t) \in \mathbb{R}^n \times \mathbb{R} : \ t \geq f(x)\}.$$

Then

$$f^*(y) = \sup_x \{x^\top y - f(x)\} = \sup_{(x,t)\in \mathrm{epi}(f)} \{x^\top y - t\}.$$

Equivalent geometric statement

The quantity $f^*(y)$ is the support value of the epigraph in the direction $(y, -1)$.

Indeed, the inequality

$$x^\top y - t \leq f^*(y)$$

can be rearranged as

$$t \geq x^\top y - f^*(y).$$

So for every y, the affine function

$$x \mapsto x^\top y - f^*(y)$$

defines a hyperplane lying below $\mathrm{epi}(f)$.

The master inequality behind everything

By definition of the supremum,

$$f^*(y) = \sup_u \{u^\top y - f(u)\}.$$

Therefore, for every fixed x,

$$x^\top y - f(x) \le f^*(y).$$

Rearranging gives

$$f(x) + f^*(y) \ge x^\top y.$$

Fenchel's inequality

For all x, y,

$$f(x) + f^*(y) \ge x^\top y.$$

This is the fundamental inequality of conjugate theory.

Proof of Fenchel's inequality, step by step

Start from the definition:

$$f^*(y) = \sup_u \{u^\top y - f(u)\}.$$

Step 1: plug in a particular point

A supremum is at least as large as the value at any specific choice of u. In particular, choose $u = x$:

$$f^*(y) \geq x^\top y - f(x).$$

Step 2: move terms

Add $f(x)$ to both sides:

$$f(x) + f^*(y) \geq x^\top y.$$

Equality case

Equality holds exactly when x attains the supremum in the definition of $f^*(y)$.

Biconjugate: why $f^{**} \le f$ always

Apply Fenchel's inequality to the pair (x, y):

$$f(x) + f^*(y) \ge x^\top y.$$

Rearrange:

$$x^\top y - f^*(y) \le f(x).$$

Now take the supremum over y:

$$\sup_y \{x^\top y - f^*(y)\} \le f(x).$$

But the left-hand side is exactly $f^{**}(x)$, so

$$f^{**}(x) \le f(x) \qquad \forall x.$$

Meaning

The biconjugate is the **largest closed convex function below** f. At this stage we have only proved the "below" part.

Why should $f^{**} = f$ for closed convex functions?

If f is closed and convex, then every point on its epigraph has a supporting hyperplane.
At a fixed x_0, that supporting hyperplane has the form

$$t \geq y^\top x - \alpha$$

and it touches the graph at $(x_0, f(x_0))$.
Since it touches at x_0,

$$f(x_0) = y^\top x_0 - \alpha.$$

Since it lies below the graph everywhere,

$$y^\top x - \alpha \leq f(x) \qquad \forall x.$$

Key idea

Every supporting affine minorant of f appears inside the supremum defining f^{**}, and if one of them touches f at x_0, then $f^{**}(x_0)$ must equal $f(x_0)$.

Biconjugate theorem: statement

Fenchel-Moreau theorem

If f is closed and convex, then
$$f^{**} = f.$$

Combined with the previous slide:

$$f^{**} \le f \quad \text{always,} \qquad f^{**} = f \quad \text{for closed convex } f.$$

Interpretation

A closed convex function is completely determined by all of its affine lower bounds.

Proof of $f^{**} = f$: the easy direction

We already proved

$$f^{**}(x) \le f(x) \qquad \forall x.$$

So the remaining task is to show the opposite inequality

$$f(x) \le f^{**}(x)$$

when f is closed and convex.

Equivalent reformulation

We must show that for each x_0, there exists an affine lower bound of f that touches f at x_0.

If such a lower bound has the form

$$x \mapsto y^\top x - \alpha,$$

then

$$y^\top x - \alpha \le f(x) \quad \forall x$$

implies

$$-\alpha \le \sup_x \{y^\top x - f(x)\} = f^*(y).$$

So

$$y^\top x_0 - f^*(y) \le f^{**}(x_0).$$

If equality holds at x_0, then we are done.

Proof of $f^{**} = f$: supporting hyperplane at $(x_0, f(x_0))$

Fix x_0. Because f is closed and convex, its epigraph

$$\text{epi}(f) = \{(x, t) : t \geq f(x)\}$$

is a closed convex set.

The point $(x_0, f(x_0))$ lies on its boundary. By the supporting hyperplane theorem, there exists a nonzero vector $(y, -1)$ and a scalar a such that

$$u^\top y - s \leq a \qquad \forall (u, s) \in \text{epi}(f),$$

and equality holds at $(x_0, f(x_0))$:

$$x_0^\top y - f(x_0) = a.$$

Now for every u, the point $(u, f(u))$ lies in $\text{epi}(f)$, so

$$u^\top y - f(u) \leq a = x_0^\top y - f(x_0).$$

Taking the supremum over u,

$$f^*(y) = \sup_u \{u^\top y - f(u)\} \leq x_0^\top y - f(x_0).$$

Proof of $f^{**} = f$: finish the argument

From the previous slide,

$$f^*(y) \le x_0^\top y - f(x_0).$$

Rearrange:

$$x_0^\top y - f^*(y) \ge f(x_0).$$

Now take the supremum over all y:

$$f^{**}(x_0) = \sup_y \{x_0^\top y - f^*(y)\} \ge f(x_0).$$

But we already know

$$f^{**}(x_0) \le f(x_0).$$

Therefore

$$f^{**}(x_0) = f(x_0).$$

Since x_0 was arbitrary, we conclude

$$f^{**} = f.$$

Big picture

Closed convex functions are exactly the functions that can be recovered from their conjugates.

Subdifferential correspondence: statement

Assume f is closed and convex. Then the following are equivalent:

$$y \in \partial f(x), \qquad x \in \partial f^*(y), \qquad f(x) + f^*(y) = x^\top y.$$

Why this matters

The gradient / subgradient map of f and the gradient / subgradient map of f^* are inverses of each other.

If f is differentiable, the statement becomes

$$y = \nabla f(x) \iff x = \nabla f^*(y).$$

This is one of the most useful formulas in optimization.

Proof, direction 1: $y \in \partial f(x)$ implies Fenchel equality

By definition,

$$y \in \partial f(x) \quad \Longleftrightarrow \quad f(u) \geq f(x) + y^\top (u - x) \qquad \forall u.$$

Rearrange:

$$u^\top y - f(u) \leq x^\top y - f(x) \qquad \forall u.$$

Now take the supremum over u:

$$f^*(y) = \sup_u \{u^\top y - f(u)\} \leq x^\top y - f(x).$$

But Fenchel's inequality always gives

$$f(x) + f^*(y) \geq x^\top y.$$

Combining the two inequalities forces equality:

$$f(x) + f^*(y) = x^\top y.$$

Proof, direction 2: Fenchel equality implies $x \in \partial f^*(y)$

Assume
$$f(x) + f^*(y) = x^\top y.$$
Take any $v \in \mathbb{R}^n$. Fenchel's inequality applied to (x, v) gives
$$f(x) + f^*(v) \geq x^\top v.$$
Subtract the equality $f(x) = x^\top y - f^*(y)$ from both sides:
$$f^*(v) \geq f^*(y) + x^\top (v - y).$$
This is exactly the subgradient inequality for f^* at y:
$$x \in \partial f^*(y).$$
So we have shown
$$y \in \partial f(x) \implies f(x) + f^*(y) = x^\top y \implies x \in \partial f^*(y).$$

Proof, direction 3: symmetry gives the converse

The previous argument is symmetric in f and f^*.
Indeed, if $x \in \partial f^*(y)$, then by applying the same reasoning to the closed convex function f^*, we obtain

$$f^*(y) + f^{**}(x) = x^\top y.$$

Since f is closed and convex,

$$f^{**}(x) = f(x).$$

Hence

$$f(x) + f^*(y) = x^\top y.$$

Then the same argument as before implies

$$y \in \partial f(x).$$

Conclusion

$$y \in \partial f(x) \iff f(x) + f^*(y) = x^\top y \iff x \in \partial f^*(y).$$

Separable sum property

Suppose

$$f(u, v) = f_1(u) + f_2(v).$$

Then its conjugate separates:

$$f^*(w, z) = f_1^*(w) + f_2^*(z).$$

Proof.

By definition,

$$f^*(w, z) = \sup_{u,v}\{u^\top w + v^\top z - f_1(u) - f_2(v)\}.$$

Group the u-terms and the v-terms:

$$= \sup_{u,v}\Big[(u^\top w - f_1(u)) + (v^\top z - f_2(v))\Big].$$

Because the variables are independent,

$$\sup_{u,v}(A(u) + B(v)) = \sup_u A(u) + \sup_v B(v).$$

Therefore

$$f^*(w, z) = f_1^*(w) + f_2^*(z).$$

□

Example: indicator function and support function

Let

$$I_C(x) = \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C. \end{cases}$$

Then

$$I_C^*(y) = \sup_x \{x^\top y - I_C(x)\} = \sup_{x \in C} x^\top y.$$

Support function

$$S_C(y) = \sup_{x \in C} x^\top y.$$

Therefore

$$I_C^* = S_C.$$

Interpretation

Indicator functions encode **membership** in a set. Support functions encode **linear optimization** over that set. Conjugacy turns one viewpoint into the other.

Example: norm $\longleftrightarrow$ indicator of the dual unit ball

Let $f(x) = \|x\|$. Then

$$f^*(y) = \sup_x \{x^\top y - \|x\|\}.$$

Claim

$$f^*(y) = \begin{cases} 0, & \|y\|_* \le 1, \\ +\infty, & \|y\|_* > 1. \end{cases}$$

Case 1: $\|y\|_* \le 1$

By generalized Cauchy-Schwarz,

$$x^\top y \le \|x\| \, \|y\|_* \le \|x\| .$$

So

$$x^\top y - \|x\| \le 0 \qquad \forall x.$$

Taking $x = 0$ gives value 0, hence $f^*(y) = 0$.

Same example: the case $\|y\|_* > 1$

Assume

$$\|y\|_* > 1.$$

By definition of the dual norm, choose u with

$$\|u\| = 1, \qquad u^\top y = \|y\|_* .$$

Now take

$$x = tu, \qquad t > 0.$$

Then

$$x^\top y - \|x\| = t\, u^\top y - t\,\|u\| = t(\|y\|_* - 1) \to +\infty.$$

So the supremum is unbounded:

$$f^*(y) = +\infty.$$

Therefore

$$(\|\cdot\|)^* = I_{\{\|y\|_* \le 1\}}.$$

Example: quadratic function

Let

$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c, \qquad A \succ 0.$$

Then

$$f^*(y) = \sup_x \left\{ x^\top y - \frac{1}{2}x^\top Ax - b^\top x - c \right\}.$$

Combine the linear terms:

$$f^*(y) = \sup_x \left\{ x^\top (y - b) - \frac{1}{2}x^\top Ax \right\} - c.$$

Since $A \succ 0$, the objective is a strictly concave quadratic in x, so it has a unique maximizer.

Quadratic conjugate: compute the maximizer

Differentiate with respect to x:

$$\nabla_x \left(x^\top (y - b) - \frac{1}{2} x^\top A x \right) = (y - b) - Ax.$$

Set the gradient equal to zero:

$$(y - b) - Ax = 0 \qquad \Longrightarrow \qquad x^\star = A^{-1}(y - b).$$

Now substitute this optimizer back into the objective:

$$f^*(y) = (x^\star)^\top (y - b) - \frac{1}{2}(x^\star)^\top A x^\star - c.$$

Because $Ax^\star = y - b$,

$$(x^\star)^\top (y - b) = (x^\star)^\top A x^\star.$$

So

$$f^*(y) = \frac{1}{2}(x^\star)^\top A x^\star - c.$$

Quadratic conjugate: simplify the final expression

Using

$$x^\star = A^{-1}(y-b),$$

we get

$$(x^\star)^\top A x^\star = (y-b)^\top A^{-1}(y-b).$$

Therefore

$$f^*(y) = \frac{1}{2}(y-b)^\top A^{-1}(y-b) - c.$$

Final formula

If

$$f(x) = \frac{1}{2}x^\top A x + b^\top x + c, \qquad A \succ 0,$$

then

$$f^*(y) = \frac{1}{2}(y-b)^\top A^{-1}(y-b) - c.$$

Example: negative entropy

Consider

$$f(x) = \sum_{i=1}^{n} x_i \log x_i$$

(on the domain $x_i > 0$).
Because the function is separable,

$$f^*(y) = \sum_{i=1}^{n} \sup_{x_i > 0} \{x_i y_i - x_i \log x_i\}.$$

So it is enough to solve the scalar problem

$$\sup_{x>0} \{xy - x \log x\}.$$

Differentiate:

$$\frac{\mathrm{d}}{\mathrm{d}x}(xy - x \log x) = y - (\log x + 1).$$

Set equal to zero:

$$y - (\log x + 1) = 0 \qquad \Longrightarrow \qquad x = e^{y-1}.$$

Negative entropy: finish the calculation

Substitute

$$x^\star = e^{y-1}$$

into the scalar objective:

$$x^\star y - x^\star \log x^\star = e^{y-1} y - e^{y-1}(y-1) = e^{y-1}.$$

Hence the scalar conjugate is

$$(x \log x)^*(y) = e^{y-1}.$$

Summing coordinatewise,

$$f^*(y) = \sum_{i=1}^{n} e^{y_i - 1}.$$

Why this is useful

This pair

$$\sum_i x_i \log x_i \quad \longleftrightarrow \quad \sum_i e^{y_i - 1}$$

appears constantly in entropy regularization, mirror descent, exponential families, and KL-based optimization.

Two scalar conjugate pairs to remember

$f(x) = e^x$

$$f^*(y) = \sup_x \{xy - e^x\}.$$

Stationarity:

$$y - e^x = 0 \Rightarrow e^x = y \Rightarrow x = \log y.$$

This only makes sense for $y > 0$.
Substitute:

$$f^*(y) = y \log y - y, \qquad y > 0.$$

$f(x) = -\log x$ on $x > 0$

$$f^*(y) = \sup_{x>0} \{xy + \log x\}.$$

Stationarity:

$$y + \frac{1}{x} = 0 \Rightarrow x = -\frac{1}{y}.$$

This requires $y < 0$.
Substitute:

$$f^*(y) = -1 - \log(-y), \qquad y < 0.$$

Conjugate calculus: why rules matter

In practice, we almost never compute conjugates from scratch every time.
Instead, we build them from a handful of rules:

- separable sums,
- scalar multiplication,
- adding affine terms,
- translations,
- linear changes of variables,
- infimal convolution.

Strategy

When deriving dual problems, the goal is to rewrite the dual function in terms of known conjugates.

Rule 1: scalar multiplication

Let

$$f(x) = \alpha g(x), \qquad \alpha > 0.$$

Then

$$f^*(y) = \sup_x \{x^\top y - \alpha g(x)\}.$$

Factor out α:

$$f^*(y) = \alpha \sup_x \left\{x^\top \frac{y}{\alpha} - g(x)\right\} = \alpha g^* \left(\frac{y}{\alpha}\right).$$

Formula

$$(\alpha g)^*(y) = \alpha g^*(y/\alpha).$$

There is a closely related second rule: if

$$f(x) = \alpha g(x/\alpha),$$

then

$$f^*(y) = \alpha g^*(y).$$

That rule comes from the substitution $u = x/\alpha$.

Rule 2: adding an affine function

Suppose

$$f(x) = g(x) + a^\top x + b.$$

Then

$$f^*(y) = \sup_x \{x^\top y - g(x) - a^\top x - b\}.$$

Group the linear terms:

$$= \sup_x \{x^\top (y - a) - g(x)\} - b.$$

The supremum is exactly $g^*(y-a)$, so

$$f^*(y) = g^*(y - a) - b.$$

Formula

$$(g(x) + a^\top x + b)^*(y) = g^*(y - a) - b.$$

Rule 3: translation

Suppose

$$f(x) = g(x - b).$$

Then

$$f^*(y) = \sup_x \{x^\top y - g(x - b)\}.$$

Make the substitution

$$u = x - b \qquad \Longrightarrow \qquad x = u + b.$$

Then

$$f^*(y) = \sup_u \{(u + b)^\top y - g(u)\} = b^\top y + \sup_u \{u^\top y - g(u)\}.$$

Hence

$$f^*(y) = b^\top y + g^*(y).$$

Formula

$$(g(\cdot - b))^*(y) = b^\top y + g^*(y).$$

Rule 4: composition with an invertible linear map

Let

$$f(x) = g(Ax),$$

where A is invertible.
Then

$$f^*(y) = \sup_x \{x^\top y - g(Ax)\}.$$

Set

$$u = Ax \qquad \Longrightarrow \qquad x = A^{-1}u.$$

Then

$$x^\top y = u^\top A^{-\top} y.$$

So

$$f^*(y) = \sup_u \{u^\top A^{-\top} y - g(u)\} = g^*(A^{-\top} y).$$

Formula

$$(g \circ A)^*(y) = g^*(A^{-\top} y), \qquad A \text{ invertible.}$$

Rule 5: infimal convolution

Define

$$f(x) = \inf_{u+v=x} \{g(u) + h(v)\}.$$

This operation is called the **infimal convolution** of g and h.
We compute the conjugate:

$$f^*(y) = \sup_x \left\{ x^\top y - \inf_{u+v=x} [g(u) + h(v)] \right\}.$$

Replace x by $u + v$:

$$f^*(y) = \sup_{u,v} \{(u + v)^\top y - g(u) - h(v)\}.$$

Separate the two variables:

$$= \sup_u \{u^\top y - g(u)\} + \sup_v \{v^\top y - h(v)\}.$$

Therefore

$$f^*(y) = g^*(y) + h^*(y).$$

Why conjugates know about curvature

Strong convexity says the primal function has a uniform quadratic lower bound:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|_2^2 .$$

Smoothness says the gradient does not move too fast:

$$\|\nabla f(y) - \nabla f(x)\|_2 \leq L \|y - x\|_2 .$$

Conjugate duality of curvature

Roughly speaking,

$$f \text{ strongly convex} \quad \Longleftrightarrow \quad f^* \text{ smooth.}$$

This is one of the most important reasons conjugates matter in optimization algorithms.

Strong convexity implies unique maximizers in the conjugate

Assume f is m-strongly convex with $m > 0$.
Fix y. Consider the function

$$x \mapsto x^\top y - f(x).$$

Since $x^\top y$ is affine and $-f(x)$ is *strictly concave* when f is strongly convex, the whole expression is strongly concave in x.

Consequence

The supremum

$$f^*(y) = \sup_x \{x^\top y - f(x)\}$$

is attained at a **unique** point.

So it makes sense to define

$$x(y) = \arg\max_x \{x^\top y - f(x)\}.$$

Later we will show that

$$x(y) = \nabla f^*(y).$$

The argmax formula for $\nabla f^*(y)$

Let

$$x^\star(y) = \arg\max_x \{x^\top y - f(x)\}.$$

Because the maximizer is unique, we can differentiate the objective with respect to x and set it to zero:

$$y - \nabla f(x^\star(y)) = 0.$$

Thus

$$y = \nabla f(x^\star(y)).$$

By the subdifferential correspondence,

$$x^\star(y) \in \partial f^*(y).$$

Since the argmax is unique, the subgradient is unique, and therefore f^* is differentiable at y with

$$\nabla f^*(y) = x^\star(y) = \arg\max_x \{x^\top y - f(x)\}.$$

Key identity

$$\nabla f^* = (\nabla f)^{-1}.$$

At least under the regularity assumptions we are using here.

Proof that ∇f^* is $(1/m)$-Lipschitz

Take any y_1, y_2 and define

$$x_1 = \nabla f^*(y_1), \qquad x_2 = \nabla f^*(y_2).$$

Then by the previous slide,

$$y_1 = \nabla f(x_1), \qquad y_2 = \nabla f(x_2).$$

Strong convexity of f implies strong monotonicity of its gradient:

$$\langle \nabla f(x_1) - \nabla f(x_2), x_1 - x_2 \rangle \geq m \left\| x_1 - x_2 \right\|_2^2 .$$

Substitute $\nabla f(x_1) = y_1$ and $\nabla f(x_2) = y_2$:

$$\langle y_1 - y_2, x_1 - x_2 \rangle \geq m \left\| x_1 - x_2 \right\|_2^2 .$$

Now apply Cauchy-Schwarz:

$$\langle y_1 - y_2, x_1 - x_2 \rangle \leq \left\| y_1 - y_2 \right\|_2 \left\| x_1 - x_2 \right\|_2 .$$

Therefore

$$m \left\| x_1 - x_2 \right\|_2^2 \leq \left\| y_1 - y_2 \right\|_2 \left\| x_1 - x_2 \right\|_2 .$$

If $x_1 \neq x_2$, divide by $\left\| x_1 - x_2 \right\|_2$:

$$\left\| x_1 - x_2 \right\|_2 \leq \frac{1}{m} \left\| y_1 - y_2 \right\|_2 .$$

So

$$\left\| \nabla f^*(y_1) - \nabla f^*(y_2) \right\|_2 \leq \frac{1}{m} \left\| y_1 - y_2 \right\|_2 .$$

The theorem: conjugate of a strongly convex function

Theorem

If f is m-strongly convex, then

- f^* is differentiable,
- $$\nabla f^*(y) = \arg\max_x \{x^\top y - f(x)\},$$
- ∇f^* is $(1/m)$-Lipschitz, i.e. f^* is $(1/m)$-smooth.

So curvature in the primal becomes gradient regularity in the dual.

Memory aid

$$m\text{-strongly convex } f \quad \Longrightarrow \quad (1/m)\text{-smooth } f^*.$$

A simple example: $f(x) = x^2$

Take

$$f(x) = x^2.$$

Then

$$f''(x) = 2,$$

so f is 2-strongly convex.
From our earlier computation,

$$f^*(y) = \frac{y^2}{4}.$$

Differentiate:

$$\nabla f^*(y) = \frac{y}{2}, \qquad (\nabla f^*)'(y) = \frac{1}{2}.$$

Hence ∇f^* is $(1/2)$-Lipschitz.

Check the constants

Since $m = 2$, the theorem predicts smoothness constant

$$\frac{1}{m} = \frac{1}{2},$$

which matches exactly.

Condition number is preserved under conjugacy

Suppose f is both

$$m\text{-strongly convex} \qquad \text{and} \qquad L\text{-smooth.}$$

Then, under standard regularity assumptions, the conjugate satisfies

$$f^* \text{ is } \frac{1}{L}\text{-strongly convex} \qquad \text{and} \qquad f^* \text{ is } \frac{1}{m}\text{-smooth.}$$

Therefore the condition number is preserved:

$$\kappa(f) = \frac{L}{m}, \qquad \kappa(f^*) = \frac{1/m}{1/L} = \frac{L}{m}.$$

Interpretation

Passing to the conjugate inverts the curvature constants but keeps their ratio.

The key identity used in dual derivations

By definition,

$$f^*(u) = \sup_x \{u^\top x - f(x)\}.$$

Negate both sides:

$$-f^*(u) = \inf_x \{f(x) - u^\top x\}.$$

This is the identity to remember

$$-f^*(u) = \inf_x \{f(x) - u^\top x\}.$$

Whenever a dual function contains an infimum of the form

$$\inf_x \{f(x) - u^\top x\},$$

we should immediately think: "this is $-f^*(u)$."

Dual of $\min_x f(x) + h(x)$

Start with

$$\min_x \; f(x) + h(x).$$

Introduce a copy variable:

$$\min_{x,y} \; f(x) + h(y) \qquad \text{s.t.} \qquad x = y.$$

The Lagrangian is

$$\mathcal{L}(x, y, \mu) = f(x) + h(y) - \mu^\top (x - y).$$

Now minimize over x and y:

$$g(\mu) = \inf_x \{f(x) - \mu^\top x\} + \inf_y \{h(y) + \mu^\top y\}.$$

Use the conjugate identity twice:

$$\inf_x \{f(x) - \mu^\top x\} = -f^*(\mu),$$

$$\inf_y \{h(y) - (-\mu)^\top y\} = -h^*(-\mu).$$

Therefore

$$g(\mu) = -f^*(\mu) - h^*(-\mu).$$

Final dual of the sum problem

From the previous slide, the dual problem is

$$\max_{\mu} \; -f^*(\mu) - h^*(-\mu).$$

Conjugate form of sum duality

$$\boxed{\min_{x}\{f(x) + h(x)\} \quad \Longleftrightarrow \quad \max_{\mu}\{-f^*(\mu) - h^*(-\mu)\}.}$$

This is the cleanest abstract form of many dual derivations.

Why it is useful

Once you know the conjugates of f and h, the dual becomes almost automatic.

Example: norm regularization

Take

$$h(x) = \|x\|.$$

We already know

$$h^*(\mu) = I_{\{\|\mu\|_* \leq 1\}}.$$

So the primal

$$\min_x \; f(x) + \|x\|$$

has dual

$$\max_\mu \; -f^*(\mu) - I_{\{\|-\mu\|_* \leq 1\}}.$$

Because $\|-\mu\|_* = \|\mu\|_*$, this simplifies to

$$\max_\mu \; -f^*(\mu) \qquad \text{s.t.} \qquad \|\mu\|_* \leq 1.$$

Lesson

A norm penalty in the primal becomes a dual-norm ball constraint in the dual.

Example: constraint set via an indicator

Take

$$h(x) = I_C(x).$$

Then

$$h^*(u) = S_C(u) = \sup_{x \in C} x^\top u.$$

So the primal

$$\min_x \; f(x) + I_C(x)$$

has dual

$$\max_\mu \; -f^*(\mu) - S_C(-\mu).$$

Interpretation

The hard constraint $x \in C$ in the primal becomes the support function of C in the dual.

This is another manifestation of the indicator-support correspondence.

Dual of the dual is the primal: two ingredients

Write the primal in abstract form:

$$\text{(P)} \qquad \min_x \; f(x) + I_C(x).$$

Its dual is

$$\text{(D)} \qquad \max_u \; -f^*(u) - S_C(-u).$$

To dualize again, we need two facts:

1. For closed convex f,

$$f^{**} = f.$$

2. For closed convex C,

$$I_C^* = S_C \qquad \text{and} \qquad S_C^* = I_C.$$

Once these are in place, taking the dual twice cancels the conjugations.

Proof that $S_C^* = I_C$: case 1, $x \in C$

Recall

$$S_C(y) = \sup_{u \in C} u^\top y.$$

We compute

$$S_C^*(x) = \sup_y \{y^\top x - S_C(y)\}.$$

Assume first that $x \in C$.
Since $S_C(y)$ is the supremum of $u^\top y$ over $u \in C$, it is at least as large as the value at $u = x$:

$$S_C(y) \geq x^\top y.$$

Therefore

$$y^\top x - S_C(y) \leq 0 \qquad \forall y.$$

Taking $y = 0$ gives value 0, so the supremum is exactly 0:

$$S_C^*(x) = 0 = I_C(x).$$

Proof that $S_C^* = I_C$: case 2, $x \notin C$

Now assume $x \notin C$ and that C is closed and convex.
By the strict separating hyperplane theorem, there exists some w such that

$$w^\top x > \sup_{u \in C} w^\top u = S_C(w).$$

Now scale w by $t > 0$:

$$(tw)^\top x - S_C(tw) = tw^\top x - tS_C(w) = t\big(w^\top x - S_C(w)\big).$$

The quantity in parentheses is strictly positive, so letting $t \to \infty$ gives

$$(tw)^\top x - S_C(tw) \to +\infty.$$

Hence

$$S_C^*(x) = +\infty = I_C(x).$$

Conclusion

$$S_C^* = I_C.$$

Dualizing twice recovers the primal

Start from

$$(\mathrm{D}) \qquad \max_u \{-f^*(u) - S_C(-u)\}.$$

Equivalently,

$$(\mathrm{D}) \qquad -\min_u \{f^*(u) + S_C(-u)\}.$$

Apply the same sum-duality formula to the minimization problem in u:

$$(\mathrm{DD}) \qquad \min_w \{(f^*)^*(w) + (S_C(-\cdot))^*(-w)\}.$$

Now simplify each term:

$$(f^*)^* = f$$

for closed convex f, and

$$(S_C(-\cdot))^*(-w) = I_C(w)$$

by the indicator-support correspondence plus the translation/sign rule.
So

$$(\mathrm{DD}) \qquad \min_w \{f(w) + I_C(w)\},$$

which is exactly the primal.

Linear maps move across the dual

Consider

$$\min_x \; f(x) + g(Ax).$$

Introduce an auxiliary variable $z = Ax$:

$$\min_{x,z} \; f(x) + g(z) \qquad \text{s.t.} \qquad z = Ax.$$

The Lagrangian is

$$\mathcal{L}(x, z, u) = f(x) + g(z) + u^\top (Ax - z).$$

Minimizing over x and z gives

$$g_D(u) = \inf_x \{f(x) + u^\top Ax\} + \inf_z \{g(z) - u^\top z\}.$$

Recognize the conjugates:

$$\inf_x \{f(x) - (-A^\top u)^\top x\} = -f^*(-A^\top u),$$

$$\inf_z \{g(z) - u^\top z\} = -g^*(u).$$

So the dual is

$$\max_u \; -f^*(-A^\top u) - g^*(u).$$

Why this “linear-map shift” is practically important

The primal

$$f(x) + g(Ax)$$

contains the matrix A inside the argument of g.
In the dual, the same matrix appears as

$$A^\top u$$

inside the conjugate of f.

Pattern

$$\boxed{\text{linear map in the primal argument} \quad \Longleftrightarrow \quad \text{adjoint linear map in the dual variable.}}$$

This pattern is the algebraic backbone of many primal-dual algorithms, including splitting methods and operator-based formulations.

Why add an equality constraint that was not there before?

Suppose we want to dualize

$$\min_x \; f(Ax + b).$$

If we write the Lagrangian directly, there are *no constraints*, so the dual function is just a constant:

$$g = \inf_x f(Ax + b).$$

That does not expose any useful dual structure.

Trick

Introduce an auxiliary variable

$$y = Ax + b$$

and rewrite the problem as

$$\min_{x,y} \; f(y) \qquad \text{s.t.} \qquad Ax + b = y.$$

Now the equality constraint creates a multiplier, and conjugates can enter.

Deriving the dual of $\min_x f(Ax + b)$

The reformulated primal is

$$\min_{x,y} \; f(y) \qquad \text{s.t.} \qquad Ax + b - y = 0.$$

Its Lagrangian is

$$\mathcal{L}(x, y, \mu) = f(y) + \mu^\top (Ax + b - y).$$

Now minimize over x and y separately.

Minimization over x

The term involving x is

$$\mu^\top Ax = x^\top A^\top \mu.$$

If $A^\top \mu \neq 0$, this linear term is unbounded below over $x \in \mathbb{R}^n$, so the infimum is $-\infty$. If $A^\top \mu = 0$, the x-term disappears.

So dual-feasible multipliers must satisfy

$$A^\top \mu = 0.$$

Finish the derivation of $\min_x f(Ax+b)$

Assume now that

$$A^\top \mu = 0.$$

Then the dual function becomes

$$g(\mu) = \mu^\top b + \inf_y \{f(y) - \mu^\top y\}.$$

Apply the conjugate identity:

$$\inf_y \{f(y) - \mu^\top y\} = -f^*(\mu).$$

Therefore

$$g(\mu) = b^\top \mu - f^*(\mu) \qquad \text{if } A^\top \mu = 0,$$

and $g(\mu) = -\infty$ otherwise.
So the dual problem is

$$\max_\mu \; b^\top \mu - f^*(\mu) \qquad \text{s.t.} \qquad A^\top \mu = 0.$$

Example: norm minimization $\min_x \|Ax - b\|$

Use the previous template with

$$f(y) = \|y\| .$$

We already know

$$f^*(\mu) = I_{\{\|\mu\|_* \leq 1\}}.$$

Therefore the dual becomes

$$\max_{\mu} \; b^\top \mu - I_{\{\|\mu\|_* \leq 1\}}(\mu) \qquad \text{s.t.} \qquad A^\top \mu = 0.$$

Equivalently,

$$\max_{\mu} \; b^\top \mu \qquad \text{s.t.} \qquad \|\mu\|_* \leq 1, \quad A^\top \mu = 0.$$

Depending on sign conventions in the multiplier, you may also see

$$\max_{\mu} \; -b^\top \mu \qquad \text{s.t.} \qquad \|\mu\|_* \leq 1, \quad A^\top \mu = 0.$$

The two forms are equivalent after replacing μ by $-\mu$.

Composite constraints: general pattern

Consider

$$\min_x \; f_0(A_0x + b_0) \qquad \text{s.t.} \qquad f_i(A_ix + b_i) \le 0, \quad i = 1, \dots, m.$$

Introduce variables

$$y_i = A_ix + b_i, \qquad i = 0, 1, \dots, m.$$

Then the problem becomes

$$\min_{x, y_0, \dots, y_m} \; f_0(y_0)$$

subject to

$$f_i(y_i) \le 0, \quad i = 1, \dots, m, \qquad A_ix + b_i - y_i = 0, \quad i = 0, \dots, m.$$

This separation is useful because conjugates apply directly to the y_i variables.

Composite constraints: first pass through the Lagrangian

Assign multipliers

$$\lambda_i \geq 0 \quad \text{for } f_i(y_i) \leq 0, \qquad \mu_i \quad \text{for } A_i x + b_i - y_i = 0.$$

The Lagrangian is

$$\mathcal{L} = f_0(y_0) + \sum_{i=1}^{m} \lambda_i f_i(y_i) + \sum_{i=0}^{m} \mu_i^\top (A_i x + b_i - y_i).$$

Group terms involving x:

$$\left(\sum_{i=0}^{m} A_i^\top \mu_i \right)^\top x.$$

So to avoid $-\infty$ when minimizing over x, we must have

$$\sum_{i=0}^{m} A_i^\top \mu_i = 0.$$

Once this holds, the x-variable disappears and the remaining infimum is over the y_i's.

Composite constraints: convert each infimum into a conjugate

Assume

$$\sum_{i=0}^{m} A_i^\top \mu_i = 0.$$

Then the dual function is

$$g(\lambda, \mu) = \sum_{i=0}^{m} b_i^\top \mu_i + \inf_{y_0}\{f_0(y_0) - \mu_0^\top y_0\} + \sum_{i=1}^{m} \inf_{y_i}\{\lambda_i f_i(y_i) - \mu_i^\top y_i\}.$$

The first infimum is

$$-f_0^*(\mu_0).$$

For the others, factor out $\lambda_i > 0$:

$$\lambda_i \left[f_i(y_i) - \left(\frac{\mu_i}{\lambda_i}\right)^\top y_i \right].$$

Hence

$$\inf_{y_i}\{\lambda_i f_i(y_i) - \mu_i^\top y_i\} = -\lambda_i f_i^*(\mu_i/\lambda_i).$$

Therefore

$$g(\lambda, \mu) = \sum_{i=0}^{m} b_i^\top \mu_i - f_0^*(\mu_0) - \sum_{i=1}^{m} \lambda_i f_i^*(\mu_i/\lambda_i).$$

Composite constraints: the final dual and the $\lambda_i = 0$ convention

The dual problem is

$$\max_{\lambda,\mu} \sum_{i=0}^{m} b_i^\top \mu_i - f_0^*(\mu_0) - \sum_{i=1}^{m} \lambda_i f_i^*(\mu_i/\lambda_i)$$

subject to

$$\lambda_i \geq 0, \qquad \sum_{i=0}^{m} A_i^\top \mu_i = 0.$$

What if $\lambda_i = 0$?

If $\lambda_i = 0$ and $\mu_i \neq 0$, then the term

$$-\mu_i^\top y_i$$

is linear in y_i and can be driven to $-\infty$.
So the natural convention is

$$\lambda_i f_i^*(\mu_i/\lambda_i) = \begin{cases} +\infty, & \lambda_i = 0,\ \mu_i \neq 0, \\ 0, & \lambda_i = 0,\ \mu_i = 0. \end{cases}$$

Two related problems: $\|Ax - b\|$ vs. $\frac{1}{2}\|Ax - b\|^2$

These two primals look similar:

$$\min_x \|Ax - b\|, \qquad \min_x \frac{1}{2}\|Ax - b\|^2.$$

But their duals look very different.

- For $\|Ax - b\|$, the conjugate of the norm is an indicator:

  $$(\|\cdot\|)^* = I_{\{\|\mu\|_* \leq 1\}}.$$

  So the dual gets a **constraint**

  $$\|\mu\|_* \leq 1.$$

- For $\frac{1}{2}\|Ax - b\|^2$, the conjugate becomes

  $$\left(\frac{1}{2}\|\cdot\|^2\right)^*(\mu) = \frac{1}{2}\|\mu\|_*^2.$$

  So the dual gets a **quadratic penalty**.

Conjugacy tells us exactly why the dual structure changes.

Conjugate of $\frac{1}{2}\|y\|^2$: Euclidean case

First take the Euclidean norm. Let

$$f(y) = \frac{1}{2}\|y\|_2^2 .$$

Then

$$f^*(\mu) = \sup_y \left\{ \mu^\top y - \frac{1}{2}\|y\|_2^2 \right\} .$$

Differentiate with respect to y:

$$\mu - y = 0 \qquad \Longrightarrow \qquad y^\star = \mu.$$

Substitute back:

$$f^*(\mu) = \mu^\top \mu - \frac{1}{2}\|\mu\|_2^2 = \frac{1}{2}\|\mu\|_2^2 .$$

So in Euclidean geometry,

$$\left(\frac{1}{2}\|\cdot\|_2^2\right)^* = \frac{1}{2}\|\cdot\|_2^2 .$$

Conjugate of $\frac{1}{2}\|y\|^2$: general norm

Now let $\|\cdot\|$ be any norm. Consider

$$f^*(\mu) = \sup_y \left\{ \mu^\top y - \frac{1}{2}\|y\|^2 \right\}.$$

By generalized Cauchy-Schwarz,

$$\mu^\top y \leq \|\mu\|_* \|y\|.$$

Therefore for every y,

$$\mu^\top y - \frac{1}{2}\|y\|^2 \leq \|\mu\|_* \|y\| - \frac{1}{2}\|y\|^2.$$

Set

$$r = \|y\| \geq 0.$$

Then the right side is bounded by

$$\sup_{r \geq 0} \left\{ \|\mu\|_* r - \frac{1}{2} r^2 \right\}.$$

This is a scalar concave quadratic with maximizer

$$r^\star = \|\mu\|_*,$$

and optimal value

$$\frac{1}{2}\|\mu\|_*^2.$$

Why the upper bound is attained

From the previous slide we know

$$f^*(\mu) \le \frac{1}{2} \|\mu\|_*^2 .$$

To prove equality, choose a vector u with

$$\|u\| = 1, \qquad \mu^\top u = \|\mu\|_* .$$

(One can justify the existence of such a u by compactness of the unit ball in finite dimensions.)
Now set

$$y^\star = \|\mu\|_* \, u.$$

Then

$$\|y^\star\| = \|\mu\|_* , \qquad \mu^\top y^\star = \|\mu\|_*^2 .$$

So

$$\mu^\top y^\star - \frac{1}{2} \|y^\star\|^2 = \|\mu\|_*^2 - \frac{1}{2} \|\mu\|_*^2 = \frac{1}{2} \|\mu\|_*^2 .$$

Hence

$$\left(\frac{1}{2} \|\cdot\|^2\right)^* (\mu) = \frac{1}{2} \|\mu\|_*^2 .$$

Duality gap as a certificate

Suppose x is primal feasible and (λ, μ) is dual feasible.
Then weak duality gives

$$g(\lambda, \mu) \leq p^\star \leq f(x).$$

So the computable quantity

$$f(x) - g(\lambda, \mu)$$

upper-bounds the true primal suboptimality:

$$f(x) - p^\star \leq f(x) - g(\lambda, \mu).$$

Practical interpretation

A small primal-dual gap certifies that x is close to optimal.

From the conjugate viewpoint, this is just a dressed-up version of Fenchel's inequality:

$$f(x) + f^*(y) \geq x^\top y.$$

At optimality, the inequality becomes equality.

Primal recovery from the dual solution

When strong duality holds, we can recover the primal solution by minimizing the Lagrangian at the dual optimum:

$$x^\star \in \arg\min_x \mathcal{L}(x, \lambda^\star, \mu^\star).$$

From the conjugate viewpoint, the same statement becomes

$$x^\star = \nabla f^*(u^\star)$$

whenever f is strongly convex and $u^\star$ is the relevant dual variable.

Why this is powerful

The dual problem may be easier to solve numerically, while the primal solution can be reconstructed afterward using the conjugate gradient map.

Structural correspondences revealed by conjugates

The conjugate framework makes several primal-dual patterns almost unavoidable:

1. **Number of dual variables = number of primal constraints.** Each constraint introduces a multiplier.
2. **Norm in the primal = dual norm in the dual.** This comes from

$$(\|\cdot\|)^* = I_{\{\|\cdot\|_* \leq 1\}}.$$

3. **Linear maps move to adjoints.** If Ax appears in the primal, then $A^\top$ appears in the dual.
4. **Curvature swaps.** Strong convexity of f becomes smoothness of f^*.

Takeaway

Conjugacy is not just a computational trick. It is the organizing principle behind these correspondences.

Definition of the dual norm

Dual norm

Given a norm $\|\cdot\|$, its dual norm is

$$\|z\|_* = \sup_{\|x\| \leq 1} x^\top z.$$

This is exactly the support function of the primal unit ball.

- For vector ℓ_p norms, the dual is the ℓ_q norm where

$$\frac{1}{p} + \frac{1}{q} = 1.$$

- For matrices, the operator norm and nuclear norm are dual.

Generalized Cauchy-Schwarz inequality

From the definition of the dual norm,

$$\|z\|_* = \sup_{\|u\| \leq 1} u^\top z.$$

Take any nonzero x. Write

$$u = \frac{x}{\|x\|}.$$

Then $\|u\| = 1$, so

$$u^\top z \leq \|z\|_* .$$

Multiply both sides by $\|x\|$:

$$x^\top z \leq \|x\| \, \|z\|_* .$$

Applying the same argument to $-x$ gives

$$|x^\top z| \leq \|x\| \, \|z\|_* .$$

Generalized Cauchy-Schwarz

$$| \langle x, z \rangle | \leq \|x\| \, \|z\|_* .$$

The dual of the dual norm

A beautiful involution property holds:

$$(\|\cdot\|_*)_* = \|\cdot\|\,.$$

We now prove it in the same style as earlier duality arguments.
Consider the trivial constrained problem

$$\min_y \|y\| \qquad \text{s.t.} \qquad y = x.$$

Its optimal value is obviously

$$p^\star = \|x\|\,.$$

We will compute its dual and show that the dual optimal value is exactly

$$(\|x\|_*)_*.$$

Then strong duality gives the result.

Proof of $(\|\cdot\|_*)_* = \|\cdot\|$: dual function setup

The Lagrangian is

$$\mathcal{L}(y, \mu) = \|y\| + \mu^\top (y - x).$$

So the dual function is

$$g(\mu) = -\mu^\top x + \inf_y \{\|y\| + \mu^\top y\}.$$

Everything now hinges on the inner infimum

$$\inf_y \{\|y\| + \mu^\top y\}.$$

We analyze it in two cases.

Case 1: if $\|\mu\|_* > 1$, the infimum is $-\infty$

Assume

$$\|\mu\|_* > 1.$$

By definition of the dual norm, there exists u with

$$\|u\| = 1, \qquad u^\top \mu = \|\mu\|_* .$$

Now choose

$$y = -tu, \qquad t > 0.$$

Then

$$\|y\| + \mu^\top y = t - t\, u^\top \mu = t(1 - \|\mu\|_*).$$

Since $\|\mu\|_* > 1$, the coefficient is negative, so letting $t \to \infty$ gives

$$\|y\| + \mu^\top y \to -\infty.$$

Therefore

$$\inf_y \{\|y\| + \mu^\top y\} = -\infty.$$

So such μ are not dual-feasible.

Case 2: if $\|\mu\|_* \le 1$, the infimum is 0

Assume

$$\|\mu\|_* \le 1.$$

By generalized Cauchy-Schwarz,

$$|\mu^\top y| \le \|\mu\|_* \|y\| \le \|y\| .$$

Hence

$$\mu^\top y \ge -\|y\| ,$$

and so

$$\|y\| + \mu^\top y \ge 0 \qquad \forall y.$$

Taking $y = 0$ gives value 0, so

$$\inf_y \{\|y\| + \mu^\top y\} = 0.$$

Therefore the dual function is

$$g(\mu) = \begin{cases} -\mu^\top x, & \|\mu\|_* \le 1, \\ -\infty, & \|\mu\|_* > 1. \end{cases}$$

Finish the dual-of-dual norm proof

The dual problem is

$$\max_{\mu} \; -\mu^\top x \qquad \text{s.t.} \qquad \|\mu\|_* \leq 1.$$

Replace μ by $-\mu$ to get the equivalent form

$$\max_{\mu} \; \mu^\top x \qquad \text{s.t.} \qquad \|\mu\|_* \leq 1.$$

But by definition, this optimal value is exactly

$$(\|x\|_*)_*.$$

So the dual optimal value is

$$d^\star = (\|x\|_*)_*.$$

Since the primal problem is convex and trivially feasible, strong duality holds, hence

$$\|x\| = p^\star = d^\star = (\|x\|_*)_*.$$

Conclusion

$$(\|\cdot\|_*)_* = \|\cdot\| \,.$$

Common conjugate pairs

$f(x)$	$f^*(y)$	Domain of f^*
$\frac{1}{2}\,\|x\|^2$	$\frac{1}{2}\,\|y\|_*^2$	$\mathbb{R}^n$
$\frac{1}{2}x^\top Ax + b^\top x + c\ (A \succ 0)$	$\frac{1}{2}(y-b)^\top A^{-1}(y-b) - c$	$\mathbb{R}^n$
$I_C(x)$	$S_C(y) = \sup_{x\in C} x^\top y$	$\mathbb{R}^n$
$\|x\|$	$I_{\{\|y\|_* \le 1\}}$	dual unit ball
$\sum_i x_i \log x_i$	$\sum_i e^{y_i - 1}$	$\mathbb{R}^n$
e^x	$y \log y - y$	$y > 0$
$-\log x\ (x > 0)$	$-1 - \log(-y)$	$y < 0$

Conjugate calculus rules at a glance

Operation on f	Conjugate	Condition
$\alpha g(x)$	$\alpha g^*(y/\alpha)$	$\alpha > 0$
$\alpha g(x/\alpha)$	$\alpha g^*(y)$	$\alpha > 0$
$g(x) + a^\top x + b$	$g^*(y - a) - b$	–
$g(x - b)$	$b^\top y + g^*(y)$	–
$g(Ax)$	$g^*(A^{-\top} y)$	A invertible
$g(x_1) + h(x_2)$	$g^*(y_1) + h^*(y_2)$	separable
$\inf_{u+v=x}\{g(u) + h(v)\}$	$g^*(y) + h^*(y)$	infimal convolution

Key properties to remember

1. f^* **is always convex.** It is a pointwise supremum of affine functions.
2. **Fenchel's inequality**
$$f(x) + f^*(y) \geq x^\top y.$$
3. **Biconjugate**
$$f^{**} \leq f \quad \text{always,} \qquad f^{**} = f \quad \text{for closed convex } f.$$
4. **Subgradient correspondence**
$$y \in \partial f(x) \iff x \in \partial f^*(y).$$
5. **Strong convexity / smoothness duality**
$$m\text{-strongly convex } f \implies (1/m)\text{-smooth } f^*.$$
6. **Dual norm involution**
$$(\|\cdot\|_*)_* = \|\cdot\|.$$

Big-picture summary

Conceptual message

The conjugate is the dual representation of a function, just as support functions are dual representations of sets.

- Closed convex functions are recovered by double conjugation.
- Dual problems often become short once we recognize conjugates.
- Norm penalties become dual-norm constraints.
- Affine minorants, Fenchel equality, and supporting hyperplanes are all the same picture viewed from different angles.
- Many primal-dual algorithms can be understood as moving between f and f^*.

Looking ahead

Next: Newton's method, where curvature information is used algorithmically rather than just geometrically.